

1. Question kya keh raha hai?

Tumhare paas array hai:

$a = [6, 4, 2, 1]$

Tumhe ise **non-decreasing order** mein karna hai:

$[1, 2, 4, 6]$

Lekin ek condition hai:

Tum kisi bhi **subarray ko reverse** kar sakte ho, lekin uski length **maximum k** honi chahiye.

2. Reverse ka matlab?

Agar:

$[1, 2, 3, 4]$

Tum 2 se 4 tak reverse karo:

$[1, 4, 3, 2]$

Yaani selected portion ulta ho jayega.

3. Ab main trick 🔥

Is problem ka pura answer **sirf k par depend karta hai**.

Case 1: $k == 1$

Agar:

$k = 1$

Toh hum sirf **1 element** ka subarray reverse kar sakte hain.

Example:

$[3, 1]$

Length 1 ko reverse karne se:

$[3] \rightarrow [3]$

$[1] \rightarrow [1]$

Kuch change hi nahi hoga.

Therefore:

Agar $k == 1$, array **already sorted** hona chahiye.

[1, 2, 3] $\rightarrow$ YES

[3, 1, 2] $\rightarrow$ NO

4. Case 2: $k \geq 2$

Yahan important observation hai:

Agar $k \geq 2$, toh hum **2 adjacent elements** ko reverse kar sakte hain.

For example:

[3, 1, 2]

[3,1] reverse:

[1,3,2]

Phir [3,2] reverse:

[1,2,3]

Humne basically adjacent elements ko swap kar diya.

Adjacent swap:

[a, b]

reverse karne par:

[b, a]

Ye exactly **swap** hai.

5. Adjacent swap itna important kyun hai?

Because agar tum kisi bhi **do adjacent elements ko swap** kar sakte ho, toh tum **kisi bhi array ko sort** kar sakte ho.

Example:

[3, 2, 1]

Swap first two:

[2, 3, 1]

Swap last two:

[2, 1, 3]

Swap first two:

[1, 2, 3]

Sorted 

Isliye:

$k \geq 2 \rightarrow \text{ALWAYS YES}$

Chahe array kitna bhi unsorted ho.

6. Example se samjho

Example 1

$n = 3$

$k = 2$

[1, 2, 3]

Already sorted hai.

YES

Example 2

$n = 4$

$k = 4$

[6, 4, 2, 1]

$k = 4$, toh hum poora array reverse kar sakte hain:

[6, 4, 2, 1]

reverse:

[1, 2, 4, 6]

Sorted 

YES

Example 3

$n = 4$

$k = 3$

[10, 3, 830, 14]

$k \geq 2$, therefore hum length 2 reverse kar sakte hain.

Start:

[10, 3, 830, 14]

First 2 reverse:

[3, 10, 830, 14]

Last 2 reverse:

[3, 10, 14, 830]

Sorted 

So:

YES

Example 4 — Important

$n = 2$

$k = 1$

[3, 1]

Sirf length 1 reverse kar sakte hain.

[3] $\rightarrow$ [3]

[1] $\rightarrow$ [1]

Array change nahi hoga.

Array sorted nahi hai.

Therefore:

NO

7. Toh actual logic kya hai?

Bas **2 conditions**:

if $k == 1$:

check whether array is already sorted

else:

YES

Ya aur simple:

$k > 1 \rightarrow \text{YES}$

$k == 1 \rightarrow$ array sorted hai toh YES, warna NO

8. Java Code

```
import java.util.*;
```

```
class Solution {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        int t = sc.nextInt();
```

```
        while (t-- > 0) {
```

```
            int n = sc.nextInt();
```

```
            int k = sc.nextInt();
```

```
int[] a = new int[n];

for (int i = 0; i < n; i++) {
    a[i] = sc.nextInt();
}

// If k >= 2, we can swap adjacent elements
if (k >= 2) {
    System.out.println("YES");
}
else {
    // k == 1

    boolean sorted = true;

    for (int i = 0; i < n - 1; i++) {

        if (a[i] > a[i + 1]) {
            sorted = false;
            break;
        }
    }

    System.out.println(sorted ? "YES" : "NO");
}
}
}
```

9. $k == 1$ mein ye condition kyun?

if ($a[i] > a[i + 1]$)

Non-decreasing ka matlab:

$a[i] \leq a[i+1]$

hona chahiye.

Example:

[1, 2, 2, 5]

Check:

$1 \leq 2$ ✓

$2 \leq 2$ ✓

$2 \leq 5$ ✓

Sorted.

But:

[1, 5, 3, 7]

Check:

$1 \leq 5$ ✓

$5 \leq 3$ ✗

Yahin pata chal gaya array sorted nahi hai.

10. Ek line mein pura question 🧠

$k \geq 2$ hai toh length-2 reversal se adjacent elements swap kar sakte ho, aur adjacent swaps se koi bhi array sort ho sakta hai → YES.

$k == 1$ hai toh koi actual change possible nahi → sirf already sorted array ke liye YES.

Final formula:

$k == 1$ && array sorted → YES

$k == 1$ && array not sorted → NO

$k \geq 2$ → YES

Is problem mein sorting karne ki zarurat hi nahi hai. Bas k check karo aur agar $k=1$ hai toh array sorted hai ya nahi check karo.

QUESTION---→

A. Halloumi Boxes

time limit per test 1 second

memory limit per test 256 megabytes

Theofanis is busy after his last contest, as now, he has to deliver many halloumis all over the world. He stored them inside n

boxes and each of which has some number a_i

written on it.

He wants to sort them in non-decreasing order based on their number, however, his machine works in a strange way. It can only reverse any subarray†

of boxes with length at most k

.

Find if it's possible to sort the boxes using any number of reverses.

†

Reversing a subarray means choosing two indices i

and j

(where $1 \leq i \leq j \leq n$

) and changing the array $a_1, a_2, \dots, a_n$

to $a_1, a_2, \dots, a_{i-1}, a_j, a_{j-1}, \dots, a_i, a_{j+1}, \dots, a_{n-1}, a_n$

. The length of the subarray is then $j-i+1$

.

Input

The first line contains a single integer t

$(1 \leq t \leq 100$

) — the number of test cases.

Each test case consists of two lines.

The first line of each test case contains two integers n

and k

$(1 \leq k \leq n \leq 100$

) — the number of boxes and the length of the maximum reverse that Theofanis can make.

The second line contains n

integers $a_1, a_2, \dots, a_n$

$(1 \leq a_i \leq 10^9$

) — the number written on each box.

Output

For each test case, print YES (case-insensitive), if the array can be sorted in non-decreasing order, or NO (case-insensitive) otherwise.

Example

InputCopy

5

3 2

1 2 3

3 1

9 9 9

4 4

6 4 2 1

4 3

10 3 8 30 14

2 1

3 1

OutputCopy

YES

YES

YES

YES

NO

Note

In the first two test cases, the boxes are already sorted in non-decreasing order.

In the third test case, we can reverse the whole array.

In the fourth test case, we can reverse the first two boxes and the last two boxes.

In the fifth test case, it can be shown that it's impossible to sort the boxes.

pura samjhao 0 se